

AI_TRUST_NOTES

Comprehensive Study & Reference Guide

CHAPTER 16: AI RISK MANAGEMENT

Estimated Reading Time: 15–20 minutes

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Version 1.0 | August 2026

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DISCLAIMER

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LEARNING OBJECTIVES

- Define AI risk management.
- Understand why AI risks must be managed throughout the lifecycle.
- Identify major categories of AI risk.
- Understand risk identification, assessment, mitigation, and monitoring.
- Understand the relationship between risk management and Trustworthy AI.

1. INTRODUCTION

AI systems can create risks involving safety, fairness, privacy, security, reliability, transparency, and accountability. AI risk management provides a structured approach for identifying, assessing, managing, and monitoring these risks throughout the AI lifecycle.

2. WHAT IS AI RISK MANAGEMENT?

AI risk management is the systematic process of identifying AI-related risks, analyzing their potential impact and likelihood, selecting appropriate controls, monitoring outcomes, and improving risk-management practices over time.

3. WHY AI RISK MANAGEMENT MATTERS

AI systems can affect individuals, organizations, and society. Risk management helps organizations anticipate potential harms instead of responding only after failures occur.

- Reduce potential harm.
- Support responsible AI development and deployment.
- Improve reliability and resilience.
- Strengthen governance and accountability.
- Support informed decision-making.
- Promote Trustworthy AI.

4. TYPES OF AI RISKS

AI risks can arise from technical, organizational, ethical, legal, and societal factors.

- Safety and reliability risks.
- Fairness and bias risks.
- Privacy risks.
- Security risks.
- Transparency and explainability risks.
- Accountability and governance risks.
- Robustness and resilience risks.
- Misuse and unintended-use risks.

5. AI RISK MANAGEMENT THROUGHOUT THE AI LIFECYCLE

Risk management should begin during problem definition and continue through data collection, development, evaluation, deployment, monitoring, and retirement.

- Problem Definition: identify intended purpose, stakeholders, possible harms, and risk tolerance.
- Data Collection: assess data quality, provenance, privacy, representation, and potential bias.
- Model Development: evaluate performance, robustness, security, fairness, and limitations.
- Deployment: define controls, human oversight, user information, and escalation procedures.
- Monitoring: track performance, incidents, drift, unexpected behavior, and emerging risks.
- Retirement: safely discontinue systems that are no longer appropriate or adequately controlled.

6. RISK IDENTIFICATION

Organizations should identify risks associated with intended uses, affected people, data, models, interfaces, external dependencies, and possible misuse.

7. RISK ASSESSMENT

Risk assessment generally considers the potential impact of a harmful event and the likelihood or plausibility of that event. Assessment methods should be appropriate to the system and context.

8. RISK MITIGATION

After risks are assessed, organizations can select controls to reduce likelihood, impact, or exposure.

- Improve data quality and representation.
- Use testing and validation procedures.
- Apply security controls.
- Provide meaningful human oversight.
- Limit system use where appropriate.
- Document limitations and assumptions.
- Establish incident-response procedures.

9. RISK MONITORING

AI risks can change after deployment because models, data, users, environments, and threats change. Continuous monitoring is therefore important.

- Performance degradation.
- Data or concept drift.
- Emerging bias.
- Security incidents.
- Unexpected outputs or behavior.
- Changes in intended use.

10. HUMAN OVERSIGHT AND AI RISK

Human oversight can reduce risk when humans have appropriate authority, information, competence, and time to review or intervene. Simply inserting a human into a workflow does not

automatically provide effective oversight.

11. AI RISK MANAGEMENT AND TRUSTWORTHY AI

Risk management supports several Trustworthy AI principles, including fairness, transparency, privacy, security, accountability, safety, robustness, and human oversight.

12. AI RISK MANAGEMENT CHALLENGES

Organizations may face incomplete information, rapidly changing systems, uncertain future risks, complex models, third-party dependencies, limited expertise, and differing regulatory requirements.

13. BEST PRACTICES

Organizations should integrate risk management into normal AI governance rather than treating it as a one-time assessment.

- Define clear risk ownership.
- Maintain appropriate documentation.
- Use multidisciplinary review.
- Test systems before and after deployment.
- Monitor high-impact systems continuously.
- Establish incident reporting and response.
- Review controls when systems or contexts change.
- Use recognized frameworks and standards where appropriate.

14. REAL-WORLD CONNECTION

The AI Trust case studies in this project illustrate why risk management is necessary. Recruitment AI demonstrates fairness risk; COMPAS illustrates high-impact decision risks; Microsoft Tay illustrates safety and misuse risks; generative AI hallucinations illustrate reliability risks; and automated driving systems illustrate safety and human-oversight risks.

SUMMARY

AI risk management is a continuous process for identifying, assessing, mitigating, and monitoring risks arising from AI systems. Effective risk management supports safer, fairer, more reliable, and more accountable AI.

KEY TAKEAWAYS

- AI risk management should cover the entire AI lifecycle.
- Risk identification should consider technical, human, organizational, and societal factors.
- Risk assessment should consider potential impact and likelihood or plausibility.
- Mitigation should be proportionate to the context and level of risk.
- Continuous monitoring is important because AI risks can change over time.
- AI risk management is a foundation for Trustworthy AI.

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